The documentation and process conversion measures necessary to comply with this revision shall be completed by 13 November 1993

INCH-POUND

MIL-S-19500/423B(USAF)
13 August 1993
SUPERSEDING
MIL-S-19500/423A(USAF)
28 September 1989

MILITARY SPECIFICATION
SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING
TYPES 2N5581 AND 2N5582, JANTX

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN silicon switching transistors. One level of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See 3.3.

1.3 Maximum ratings.

| $P_T$ $T_A$ = +25°C | $P_T$ $T_C$ = +25°C | $V_{CBO}$ | $V_{CEO}$ | $I_C$ | $T_{OP}$ |
|---|---|---|---|---|---|
| W | W | V dc | V dc | mA dc | °C |
| 0.5 1/ | 2.0 2/ | 75 | 50 | 800 | -55 to +200 |

1/ Derate linearly 2.86 mW/°C above $T_A$ = +25°C.
2/ Derate linearly 11.43 mW/°C above $T_C$ = +25°C.

1.4 Primary electrical characteristics.

| | $h_{FE2}$ $V_{CE}$ = 10 V dc $I_C$ = 1.0 mA dc | | $h_{FE4}$ $V_{CE}$ = 10 V dc $I_C$ = 150 mA dc 1/ | | $\|h_{fe}\|$ $V_{CE}$ = 20 V dc $I_C$ = 20 mA dc f = 100 MHz | Cobo $V_{CB}$ = 10 V dc $I_E$ = 0 100 kHz ≤ f ≤ 1 MHz | Switching | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | $t_{on}$ | $t_{off}$ | $t_{on}$+ $t_{off}$ |
| | 2N5581 | 2N5582 | 2N5581 | 2N5582 | | pF | ns | ns | ns |
| Min | 35 | 75 | 40 | 100 | 2.5 | | | | |
| Max | | | 120 | 300 | 5.0 | 8 | 35 | 300 | 18 |

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5270, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, appendix F, figure 9, T-2B.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-S-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.



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4.3 Screening (JANTX levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table II of MIL-S-19500) | Measurements |
|---|---|
| | JANTX levels |
| 11 | $I_{CBO1}$ and $h_{FE4}$ |
| 12 | See 4.3.1 |
| 13 | Subgroup 2 of table I herein; $\Delta I_{CBO1}$ = 100 percent of initial value or 5 nA dc, whichever is greater; $\Delta h_{FE4}$ = 15 percent of initial value. |

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: $T_A$ = room ambient as defined in 4.5 of the general requirements of MIL-STD-750.

$V_{CB}$ = 30 V dc; $P_T$ = 400 mW.

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JANTX and JANTXV) of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 1027 | $V_{CB}$ = 10 V dc; $P_T$ = 400 mW at $T_A$ = +30°C ±5°C. No heat sink or forced-air cooling on the devices shall be permitted. |

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition E. |
| C6 | 1026 | $V_{CB}$ = 10 V dc; $P_T$ = 400 mW at $T_A$ = +30°C ±5°C. No heat sink or forced-air cooling on device shall be permitted. |

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Breakdown voltage collector - base | 3001 | Bias condition D; $I_C$ = 10 µA dc | $V_{(BR)CBO}$ | 75 | | V dc |
| Breakdown voltage emitter - base | 3026 | Bias condition D; $I_E$ = 10 µA dc | $V_{(BR)EBO}$ | 6 | | V dc |
| Breakdown voltage collector - emitter | 3011 | Bias condition D; $I_C$ = 10 mA dc; pulsed (see 4.5.1) | $V_{(BR)CEO}$ | 50 | | V dc |
| Collector - base cutoff current | 3036 | Bias condition D; $V_{CB}$ = 60 V dc | $I_{CBO1}$ | | 10 | nA dc |
| Emitter - base cutoff current | 3061 | Bias condition D; $V_{EB}$ = 4.0 V dc | $I_{EBO1}$ | | 10 | nA dc |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 0.1 mA dc | $h_{FE1}$ | | | |
| 2N5581 | | | | 30 | | |
| 2N5582 | | | | 50 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 1.0 mA dc | $h_{FE2}$ | | | |
| 2N5581 | | | | 35 | | |
| 2N5582 | | | | 75 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 10 mA dc | $h_{FE3}$ | | | |
| 2N5581 | | | | 40 | | |
| 2N5582 | | | | 100 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 150 mA dc; pulsed (see 4.5.1) | $h_{FE4}$ | | | |
| 2N5581 | | | | 40 | 120 | |
| 2N5582 | | | | 100 | 300 | |

See footnote at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 - Continued | | | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 500 mA dc; pulsed (see 4.5.1) | $h_{FE5}$ | | | |
| 2N5581 | | | | 20 | | |
| 2N5582 | | | | 30 | | |
| Collector - emitter voltage (saturated) | 3071 | $I_C$ = 150 mA dc; $I_B$ = 15 mA dc; pulsed (see 4.5.1) | $V_{CE(sat)1}$ | | 0.3 | V dc |
| Collector - emitter voltage (saturated) | 3071 | $I_C$ = 500 mA dc; $I_B$ = 50 mA dc; pulsed (see 4.5.1) | $V_{CE(sat)2}$ | | 1.0 | V dc |
| Base - emitter saturated voltage | 3066 | Test condition A; $I_C$ = 150 mA dc; $I_B$ = 15 mA dc; pulsed (see 4.5.1) | $V_{BE(sat)1}$ | 0.6 | 1.2 | V dc |
| Base - emitter saturated voltage | 3066 | Test condition A; $I_C$ = 500 mA dc; $I_B$ = 50 mA dc; pulsed (see 4.5.1) | $V_{BE(sat)2}$ | | 2.0 | V dc |
| Subgroup 3 | | | | | | |
| High-temperature operation: | | $T_A$ = +150°C | | | | |
| Collector - base cutoff current | 3036 | Bias condition D; $V_{CB}$ = 60 V dc | $I_{CBO2}$ | | 10 | µA dc |
| Low-temperature operation: | | $T_A$ = -55°C | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 10 mA dc | $h_{FE6}$ | | | |
| 2N5581 | | | | 15 | | |
| 2N5582 | | | | 35 | | |

See footnote at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 4 | | | | | | |
| Small-signal short-circuit forward-current transfer ratio | 3206 | $V_{CE}$ = 10 V dc; $I_C$ = 1 mA dc | $h_{fe}$ | | | |
| 2N5581 | | | | 30 | | |
| 2N5582 | | | | 50 | | |
| Small-signal short-circuit forward-current transfer ratio | 3306 | $V_{CE}$ = 20 V dc; $I_C$ = 50 mA dc; f = 100 MHz | $\lvert h_{fe} \rvert$ | 2.5 | | |
| Open circuit output capacitance | 3236 | $V_{CB}$ = 10 V dc; $I_E$ = 0; 100 kHz ≤ f ≤ 1 MHz | $C_{obo}$ | | 8 | pF |
| Input capacitance (output open-circuited) | 3240 | $V_{EB}$ = 0.5 V dc; $I_C$ = 0; 100 kHz ≤ f ≤ 1 MHz | $C_{ibo}$ | | 25 | pF |
| Turn-on time | | See figure 1 | $t_{on}$ | | 35 | ns |
| Turn-off | | See figure 2 | $t_{off}$ | | 300 | ns |
| Pulse response | | See figure 3 | $t_{on} + t_{off}$ | | 18 | ns |
| Subgroups 5, 6, and 7 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan, see MIL-S-19500.



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TABLE II. Groups B and C electrical measurements.

| Step | Inspection | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| 1. | Collector - base cutoff current | 3036 | Bias condition D; $V_{CB}$ = 60 V dc | $I_{CBO1}$ | | 10 | nA dc |
| 2. | Collector - base cutoff current | 3036 | Bias condition D; $V_{CB}$ = 60 V dc | $I_{CBO3}$ | | 20 | nA dc |
| 3. | Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 150 mA dc; pulsed (see 4.5.1) | $h_{FE4}$ | | | |
| | 2N5581 | | | | 40 | 120 | |
| | 2N5582 | | | | 100 | 300 | |
| 4. | Forward-current transfer ratio | 3076 | $V_{CB}$ = 10 V dc; $I_B$ = 150 mA dc; pulsed (see 4.5.1) | $h_{FE}$ | | | |
| | 2N5581 | | | | 30 | 150 | |
| | 2N5582 | | | | 75 | 375 | |

1/ The electrical measurements for table IVb of MIL-S-19500 are as follows:

For JANTX:

a. Subgroup 2, see table II herein, steps 1 and 2.

b. Subgroups 3 and 6, see table II herein, steps 3 and 4.

2/ The electrical measurements for table V of MIL-S-19500 are as follows:

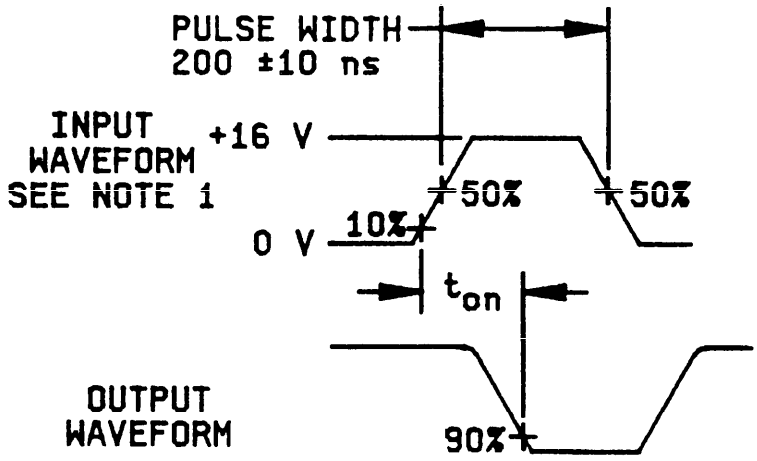
For JANTX:

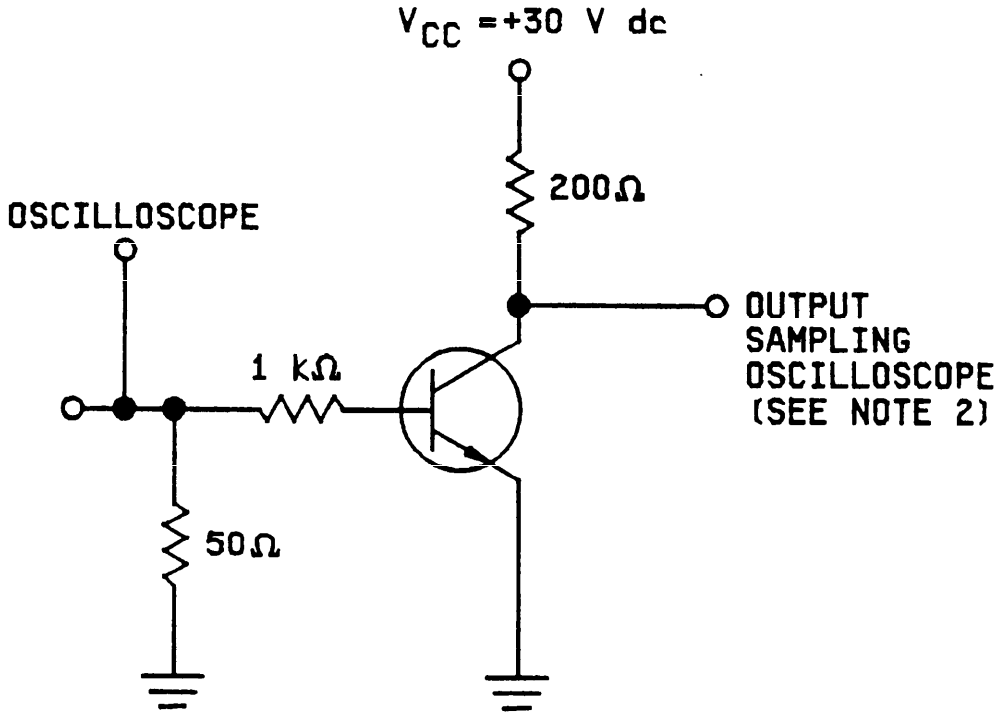
a. Subgroups 2 and 3, see table II herein, steps 1 and 2.

b. Subgroup 6, see table II herein, steps 3 and 4.



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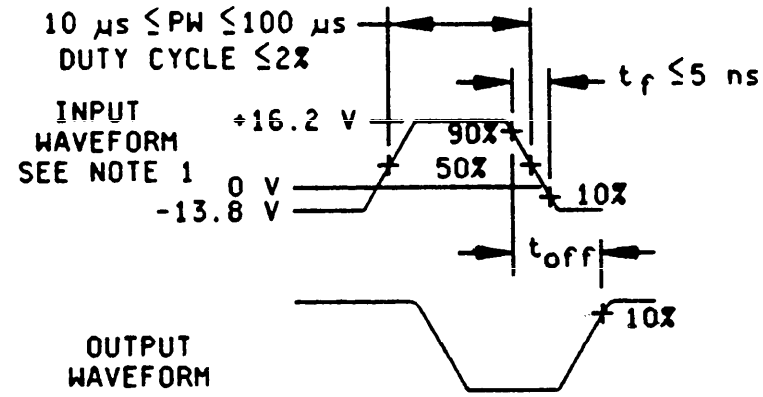



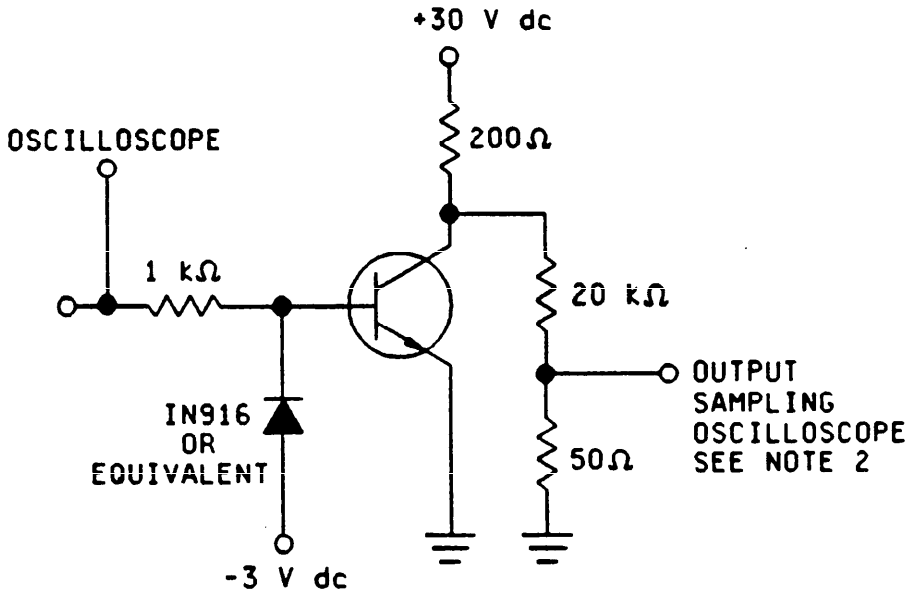

NOTES:
1. The rise time ($t_r$) and fall time ($t_f$) of the applied pulse shall be ≤ 2.0 ns; duty cycle ≤ 2 percent; generator source impedance shall be 50 ohms.
2. Output sampling oscilloscope: $Z_{in}$ ≥ 100 kΩ; $C_{in}$ ≤ 12 pF; rise time ≤ 5.0 ns.

FIGURE 1. Saturated turn-on switching time test circuit.



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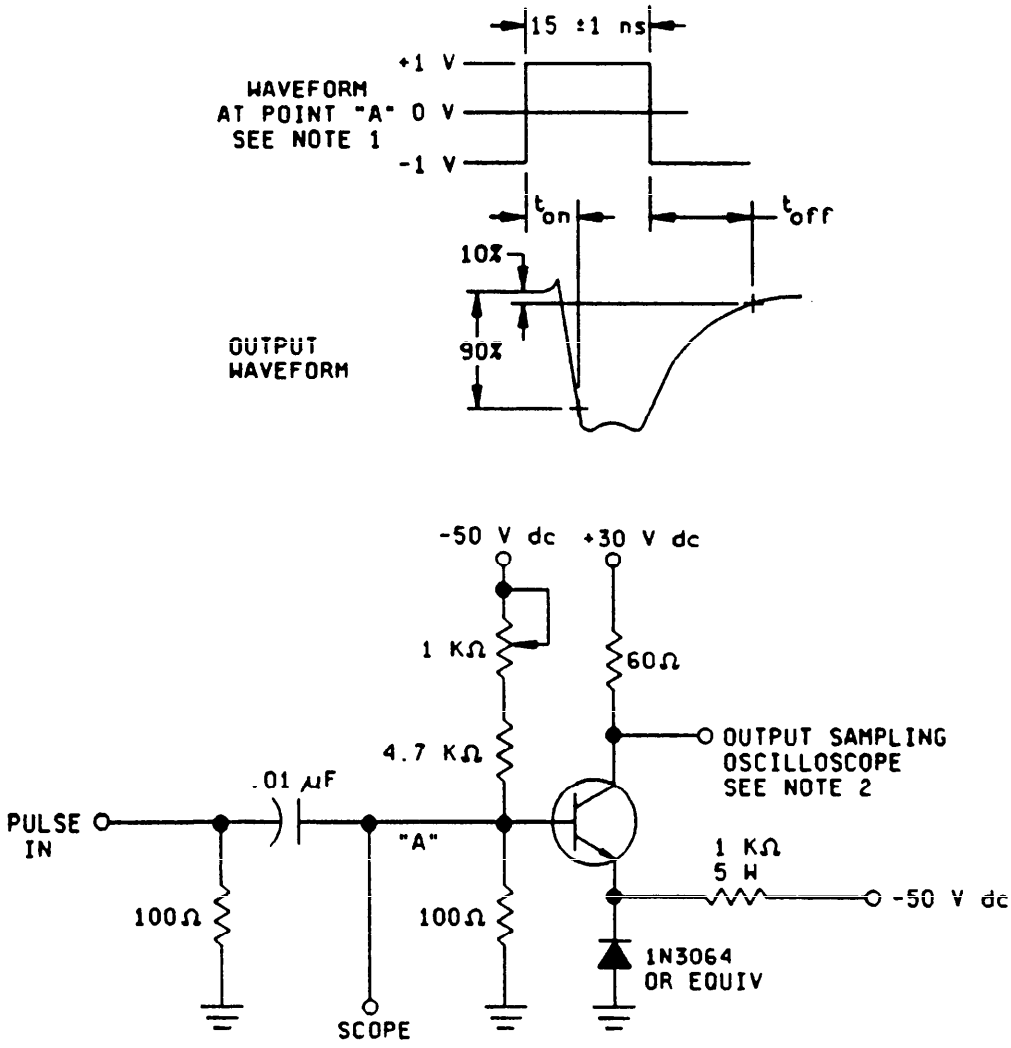




NOTES:
1. The rise time ($t_r$) and fall time ($t_f$) of the applied pulse shall be ≤ 2.0 ns; duty cycle ≤ 2 percent; generator source impedance shall be 50 ohms.
2. Output sampling oscilloscope: $Z_{in}$ ≥ 100 kΩ; $C_{in}$ ≤ 12 pF; rise time ≤ 0.2 ns.

FIGURE 2. Saturated turn-off switching time test circuit.



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NOTES:
1. The rise time ($t_r$) and fall time ($t_f$) of the applied pulse shall be ≤ 2.0 ns; duty cycle ≤ 2 percent; generator source impedance shall be 50 ohms.
2. Output sampling oscilloscope: $Z_{in}$ ≥ 100 kΩ; $C_{in}$ ≤ 12 pF; rise time ≤ 0.2 ns.

FIGURE 3. Nonsaturated turn-on switching time test circuit.



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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Lead finish (see 3.3.1).

b. Type designation and product assurance level.

6.3 Supersession information. Devices covered by this document supersede the manufacturer's and user's Part or Identifying Number (PIN). This information in no way implies that manufacturers' PINs are suitable as a substitute for the military PIN.

| Military PIN | Manufacturer's CAGE code | Manufacturer's and user's PIN |
|---|---|---|
| 2N5581JAN | 04713 | 2N5581 |
| 2N5581JTX | | 2N5581 |
| 2N5582JAN | | ST1531H |
| | | ST939H41 |
| | | SUN1036H |
| | | SUN2040H |
| 2N5582JTX | | ST1547H2 |
| | | SUN0023H10 |

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodian:
Air Force - 17

Review activities:
Air Force - 19, 85, 90
DLA - ES

User activities:
Air Force - 13, 15

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-S-19500/423B | 2. DOCUMENT DATE (YYMMDD)<br>93/12/13 |
|---|---|---|

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING
TYPE 2N5581 AND 2N5582, JANTX

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>(if applicable) | 7. DATE SUBMITTED<br>(YYMMDD) |

8. PREPARING ACTIVITY

| a. NAME<br>Alan Barone | b. TELEPHONE (Include Area Code)<br>(1) Commercial<br>513-296-6048 | (2) AUTOVON<br>986-6048 |
|---|---|---|
| c. ADDRESS (Include Zip Code)<br>ESC-ECT<br>1507 Willmington Pike<br>Dayton, OH 45444-5270 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |



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